

Retail Bank Transaction Analysis

Project Introduction

A retail bank serves customers through different banking products such as accounts, cards, and loans. The bank operates through multiple branches and handles a large number of customer transactions and loan payments.

The bank's management team wants to use data to understand its customers, account activity, transaction patterns, loan performance, and banking product usage.

You will work as a BI Analyst supporting the Retail Banking team. Your task is to understand the business, explore and clean the data, build your own data model, identify key performance indicators, create DAX measures, design interactive dashboards, and communicate meaningful business insights.

Sprint 1: Business Understanding

1.1 Company Background

Understand the retail banking business, the different banking products and services offered, and the business context behind the analysis.

1.2 Role

Assume the role of a BI Analyst supporting the Retail Banking team. Your responsibility is to use the available data to answer business questions and support banking-related decisions through interactive dashboards.

No data model or ER diagram is provided for this project. You will explore the data yourself in Sprint 2 and decide how the tables should relate to one another.

Sprint 2: Data Loading, Cleaning & Data Modeling

2.1 Import the Data

Use Get Data → Text/CSV in Power BI Desktop to load all seven tables — customers, accounts, branches, cards, loans, loan_payments, transactions — plus the provided date_table.csv.

- Review each table in Power Query Editor before loading.

- Confirm data types are set correctly: text for ID columns (they are alphanumeric codes, not numbers), date for date columns, decimal number for balance/amount/rate columns, and whole number for count-style columns such as tenure_months, employee_count, days_late, and reward_points.

Dataset Overview

*Row counts reflect the raw files as provided, including the light data-quality issues you are expected to find and resolve during Sprint 2.

Table	Shape
customers	(500,13)
accounts	(700,10)
branches	(40,7)
cards	(600,10)
loans	(300,13)
loan_payments	(2,000,11)
transactions	(5,000,10)
date_table	6,940 (2010-01-01 to 2028-12-31)

2.2 Data Cleaning in Power Query

This dataset contains a light layer of realistic data-quality issues — enough to be worth finding, not enough to fight with. Use the "Column Distribution" and "Column Profile" tools (View tab) on every categorical column before you trust it.

- Look for inconsistent casing and stray leading/trailing spaces across categorical fields such as gender, kyc_status, segment, account_type, card_type, network, loan_type, purpose, payment_method, channel, and description — a value like "Visa" and " visa " will otherwise be treated as two different categories.
- Check for blank values — a small number of customers are missing a segment. Decide how to handle this sensibly (for example, an explicit "Unspecified" category rather than silently dropping the row).
- Use Trim, Clean, and consistent capitalization to resolve what you find.

Data Note: core outcome fields — accounts.status, loans.loan_status, transactions.status, loan_payments.status, and the is_active flags — are already clean. Your cleaning effort should go into the descriptive/categorical fields listed above, not into second-guessing these.

Data Note: keep every cleaning step visible in the Applied Steps pane rather than editing values by hand. This pane is one of the most valuable habits in real BI work — it is a transparent, reproducible record of exactly what was done to the data and in what order. Take a screenshot of your Applied Steps pane for at least one table once your cleaning is complete — it is part of your deliverable.

2.3 Build the Data Model

Switch to Model view and build the relationships between the tables yourself, including the date_table. Decide which columns should connect and what each relationship's cardinality and cross-filter direction should be — this is not provided to you.

- After building each relationship, test it with one simple visual pulling a field from each of the two connected tables. If totals look inflated, duplicated, or unexpectedly missing, that is a signal to revisit the relationship — not the data.
- A table can only have one active relationship to the date_table at a time. Decide which date column on each table is the most central one for analysis (for example, transaction_date on transactions) and connect that as the active relationship; any other date column on the same table can be left as a regular column or connected inactively if you need it in a specific measure.
- Not every date column belongs in your date_table's range. branches.opening_date reaches back to 1986 and loans.maturity_date reaches out to 2053 — both fall outside date_table on purpose. Decide for yourself whether either genuinely needs a calendar relationship, or whether it is better used as a plain attribute (e.g. calculating branch age or years-to-maturity directly) instead.

Data Note: 43% of accounts (301 of 700) have no card at all — this is realistic, not an error. Any visual or relationship combining accounts and cards can show blank or missing categories for those accounts. Keep this in mind while validating your model and interpreting Sprint 5 dashboards, especially for Objective 4.5.

Sprint 3: Identify Key Performance Indicators

Before building any dashboard, identify the KPIs that actually matter to this business. A KPI is not just any number you can calculate — it is a number someone would check regularly to know whether the bank is doing well or poorly.

Consider these categories as starting points, not a fixed list:

- Customer & Account Growth — how the customer base and account portfolio are growing, and where.
- Value & Revenue — how much money the bank holds and earns, and from whom.
- Loan Performance — how reliably customers are repaying what they owe.
- Product Engagement — how deeply customers are using the bank's products beyond a single account.

For each KPI you choose, document three things: exactly how it is calculated, why it matters to the business, and what a “good” versus a “concerning” value would look like. Aim for at least 6–8 KPIs spread across the categories above before moving to Sprint 4 — these will directly feed your DAX measures and dashboards.

Sprint 4: DAX Measures and Calculated Columns

For each business objective below, create the DAX measures and/or calculated columns needed to support it. The objectives are the same ones your dashboards will be built around in Sprint 5, so plan your DAX with that end goal in mind.

Measures vs. Calculated Columns

These are not interchangeable, and choosing the wrong one is a common beginner mistake:

- A measure is calculated dynamically, based on whatever filter context is applied in a visual — use it for totals, rates, and averages (for example, Loan Delinquency Rate, Average Account Balance).
- A calculated column is computed once per row when the data refreshes, and its value does not change with filtering — use it to create a new category or flag directly on a row (for example, bucketing days_late into “On Time,” “1–30 Days Late,” and “30+ Days Late”).

Organize your measures in a dedicated Measures table. Once you have more than a handful, group them into Display Folders (for example, “Loan Performance,” “Customer,” “Card & Product”) — this is standard practice on any real BI project with a non-trivial number of measures.

Use CALCULATE, FILTER, and DIVIDE (always use DIVIDE instead of “/” so a blank denominator does not break your visual), plus basic time intelligence using the date_table for month-over-month or trend comparisons, wherever an objective calls for it.

4.1 Understand Customer Profile and Segmentation

Business Objective: The bank wants to understand its customer base and identify differences between customer groups.

Things to Consider While Analyzing:

- Compare customers across different segments.
- Look at customer demographics.
- Compare customers across cities and states.
- Examine income and credit-score differences.
- Look at customer activity and KYC status.
- Understand customer tenure with the bank.

Create the DAX measures and/or calculated columns this objective needs. The points above are starting points, not a fixed list — you decide exactly what to calculate.

4.2 Understand Account Usage and Branch Activity

Business Objective: The bank wants to understand how its accounts are being used and how account activity differs across account types and branches.

Things to Consider While Analyzing:

- Compare different account types.
- Compare account activity across customers.
- Examine account balances.
- Compare account activity across branches.
- Look at interest rates across account types.
- Identify differences between active and closed accounts.

Create the DAX measures and/or calculated columns this objective needs. The points above are starting points, not a fixed list — you decide exactly what to calculate.

4.3 Analyze Transaction Patterns

Business Objective: The bank wants to understand how customers use their accounts and how money moves through the banking system.

Things to Consider While Analyzing:

- Compare different transaction types.
- Compare transactions across different channels.
- Examine transaction amounts.
- Look at common transaction descriptions.

- Examine transaction activity over time.
- Compare transaction activity across accounts or customer groups.
- Look at how transaction activity affects account balances.

Create the DAX measures and/or calculated columns this objective needs. The points above are starting points, not a fixed list — you decide exactly what to calculate.

4.4 Evaluate Loan Performance and Repayment Behaviour

Business Objective: The bank wants to understand how its loans are performing and whether customers are repaying their loans as expected.

Things to Consider While Analyzing:

- Compare different loan types.
- Compare loans based on their purpose.
- Examine loan amounts and outstanding balances.
- Compare loan statuses.
- Identify loans with repayment delays.
- Examine penalties and late payments.
- Compare repayment behaviour across loan types or branches.
- Look at payment methods used by customers.

Create the DAX measures and/or calculated columns this objective needs. The points above are starting points, not a fixed list — you decide exactly what to calculate.

4.5 Understand Card Usage and Product Engagement

Business Objective: The bank wants to understand how customers use its card products and how card usage relates to their broader banking relationship.

Things to Consider While Analyzing:

- Compare different card types.
- Examine credit limits and outstanding balances.
- Compare active and inactive cards.
- Examine reward points.
- Compare card usage across accounts and networks.
- Identify customers using multiple banking products.
- Look at relationships between cards, accounts, and loans.

Create the DAX measures and/or calculated columns this objective needs. The points above are starting points, not a fixed list — you decide exactly what to calculate.

Sprint 5: Objective-Based Dashboard Design

For each objective, bring together the relevant KPIs (Sprint 3) and DAX measures/columns (Sprint 4) into a report page. Choose the visual types yourself — the goal is to communicate the finding clearly, not to match a specific chart type. Add slicers or filters where they would genuinely help someone explore the page.

5.1 Understand Customer Profile and Segmentation

Business Objective: The bank wants to understand its customer base and identify differences between customer groups.

Build a report page addressing this objective using the KPIs and DAX measures/columns you created for it in Sprints 3 and 4. Record your findings and business insights directly on the page.

5.2 Understand Account Usage and Branch Activity

Business Objective: The bank wants to understand how its accounts are being used and how account activity differs across account types and branches.

Build a report page addressing this objective using the KPIs and DAX measures/columns you created for it in Sprints 3 and 4. Record your findings and business insights directly on the page.

5.3 Analyze Transaction Patterns

Business Objective: The bank wants to understand how customers use their accounts and how money moves through the banking system.

Build a report page addressing this objective using the KPIs and DAX measures/columns you created for it in Sprints 3 and 4. Record your findings and business insights directly on the page.

5.4 Evaluate Loan Performance and Repayment Behaviour

Business Objective: The bank wants to understand how its loans are performing and whether customers are repaying their loans as expected.

Build a report page addressing this objective using the KPIs and DAX measures/columns you created for it in Sprints 3 and 4. Record your findings and business insights directly on the page.

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Business Objective: The bank wants to understand how customers use its card products and how card usage relates to their broader banking relationship.

Build a report page addressing this objective using the KPIs and DAX measures/columns you created for it in Sprints 3 and 4. Record your findings and business insights directly on the page.

5.6 Final Report

Build one additional page — an executive summary — bringing together the headline KPIs and the most important findings across all five objectives. Write it for a non-technical stakeholder: lead with what matters, not with every number you calculated.

Power BI Deliverable

- A Power BI Desktop file (.pbix) containing your cleaned data, your data model with the relationships you defined, your DAX measures and calculated columns organized into display folders, one dashboard page per Sprint 5 objective, and the final executive summary report page.
- A screenshot of the Power Query Applied Steps pane for at least one table, showing your cleaning steps.
- A screenshot or export of your Data Model (Model view) showing the relationships you built.

- An exported PDF of the final report (all pages), so it can be reviewed without opening Power BI Desktop.

Final Outcome

By completing this project, you should be able to move from a business requirement to a data-driven dashboard: understand the business problem, explore and clean raw data, reason your way to a data model, identify the KPIs that matter, create DAX measures, design interactive visuals, interpret results, and communicate meaningful business insights through a dashboard.